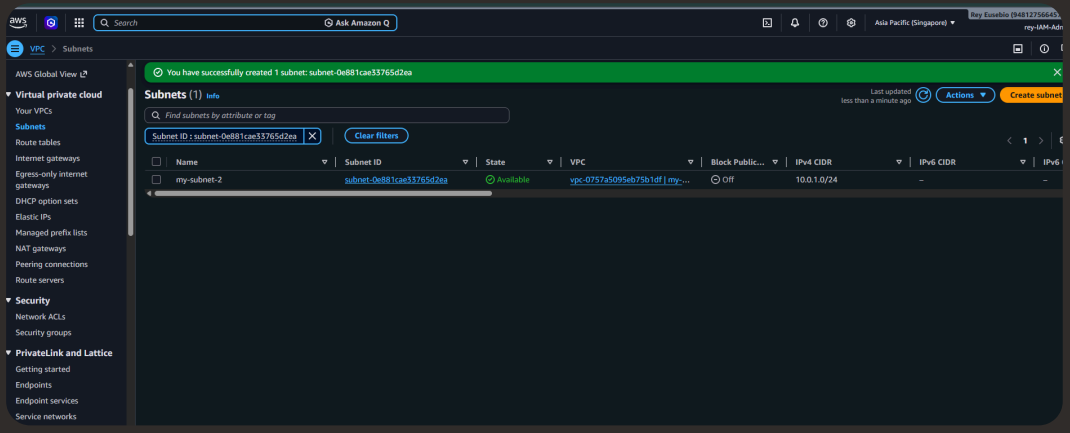
CREATING A PRIVATE SUBNET

By Reynaldo Eusebio | October 2026



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual network in AWS where I can create and manage resources such as subnets, route tables, and network security controls. It is useful because it allows me to control how my AWS resources communicate with each other and with the internet.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create a private subnet, configure a dedicated private route table, and create a private network ACL to control and restrict network traffic.

One thing I didn't expect in this project was...

One thing I didn't expect in this project is how important route tables and network ACLs are in making a subnet private and controlling its traffic.

This project took me...

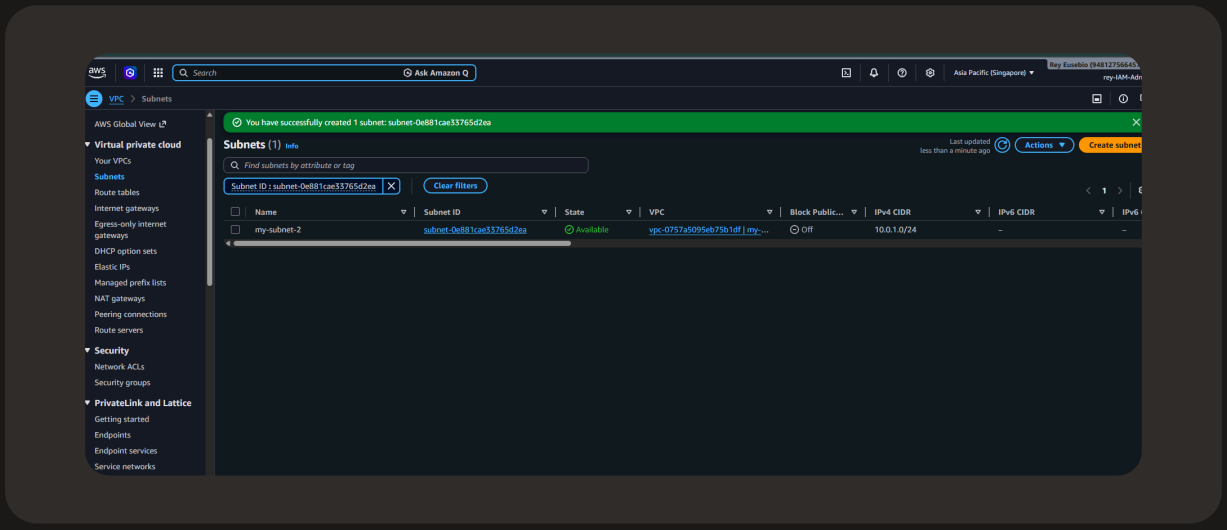
This project took me about 1 hour to complete because I spent time understanding how private subnets, route tables, and network ACLs work.

Private vs Public Subnets

A public subnet has a route to an Internet Gateway, and resources in it can be assigned a public IPv4 address. A private subnet does not have a direct route to an Internet Gateway, so resources normally use private IP addresses.

Having private subnets is useful because they keep resources isolated from direct access from the internet.

My private and public subnets cannot have the same CIDR block.

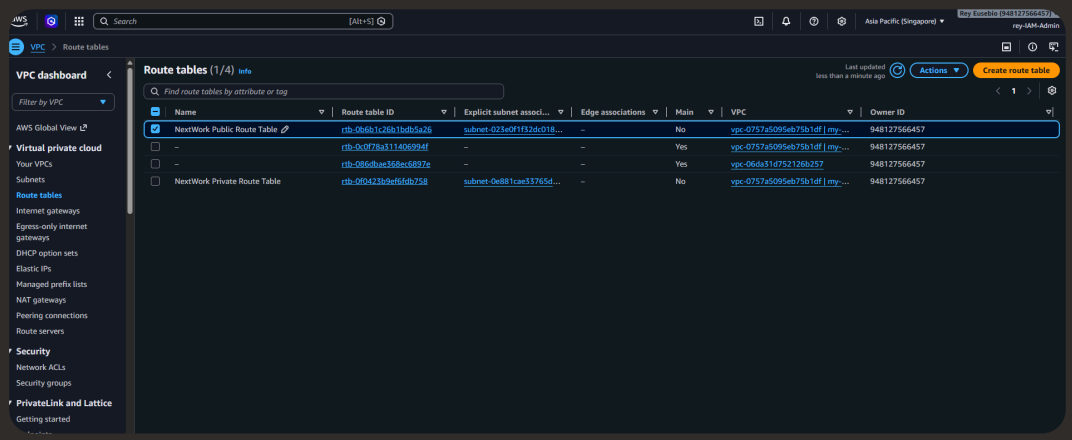


A dedicated route table

By default, my private subnet is associated with the VPC's main (default) route table.

I had to set up a new route table because the existing public route table has a route to the Internet Gateway, which would make my private subnet publicly accessible.

My private subnet's dedicated route table only has one local route that allows communication within the VPC.



A new network ACL

By default, my private subnet is associated with the VPC's default Network ACL.

I set up a dedicated network ACL for my private subnet because I want to control and restrict inbound and outbound traffic to help protect the resources in the private subnet.

My new network ACL has two simple rules — one inbound rule and one outbound rule, both denying all traffic by default.

